

## CHEMURNAUT, Russian Federation

WMO: 259410

Lat: 60.733N

Lon: 163.017E

Elev: 14

StdP: 101.16

Time Zone: 12.00 (E12)

Period: 86-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)   | (n)                   | (o)  | (p)   |
| 1             | -32.7      | -30.3 | -35.8                         | 0.1 | -32.7 | -33.3 | 0.2 | -30.1 | 18.5                  | -7.5 | 16.7 | -15.3 | 6.1                   | 160  | 0.924 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 6.2                    | 18.8            | 14.2 | 17.2 | 13.4 | 15.9 | 12.6 | 15.0                | 17.9 | 13.9 | 16.5 | 13.0 | 15.4 | 5.3                  | 160  |

| Dehumidification DP/MCDB and HR |     |      |      |     |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|------|-----|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%   |     |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP   | HR  | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d)  | (e) | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 13.5                            | 9.7 | 16.7 | 12.5 | 9.0 | 15.6 | 11.5 | 8.5 | 14.6 | 42.1          | 18.1 | 39.2 | 16.6 | 36.7 | 15.4 | 21.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 16.0          | 14.3 | 12.9 | DB                         | -34.5 | 22.7               | 2.5 | 3.0                                                | -36.3 | 24.9       | -37.8 | 26.7       | -39.2 | 28.4       | -41.0 | 30.6 | (4) |
| (5)               |      |      | WB                         | -34.3 | 17.5               | 2.4 | 2.4                                                | -36.1 | 19.2       | -37.4 | 20.6       | -38.8 | 22.0       | -40.5 | 23.7 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | -2.6  | -16.5 | -15.3 | -13.8 | -5.7  | 1.7   | 8.5   | 12.1  | 12.0  | 7.6   | -0.4  | -7.9  | -14.7 | (6)  |
| (7)  |                                                                                  | DBStd       | 11.77 | 7.04  | 7.58  | 6.71  | 4.54  | 2.77  | 2.78  | 2.30  | 1.95  | 2.81  | 4.73  | 6.81  | 6.60  | (7)  |
| (8)  |                                                                                  | HDD10.0     | 4767  | 820   | 709   | 739   | 471   | 259   | 59    | 4     | 5     | 78    | 322   | 537   | 764   | (8)  |
| (9)  |                                                                                  | HDD18.3     | 7648  | 1078  | 942   | 997   | 721   | 517   | 293   | 192   | 196   | 321   | 580   | 787   | 1023  | (9)  |
| (10) |                                                                                  | CDD10.0     | 162   | 0     | 0     | 0     | 0     | 1     | 16    | 71    | 68    | 7     | 0     | 0     | 0     | (10) |
| (11) |                                                                                  | CDD18.3     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     | (11) |
| (12) |                                                                                  | CDH23.3     | 4     | 0     | 0     | 0     | 0     | 0     | 0     | 4     | 0     | 0     | 0     | 0     | 0     | (12) |
| (13) |                                                                                  | CDH26.7     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (13) |
| (14) | Wind                                                                             | WSAvg       | 5.7   | 6.6   | 6.5   | 6.3   | 6.0   | 5.1   | 4.8   | 4.8   | 4.9   | 5.8   | 6.5   | 6.5   | (14)  |      |
| (15) | Precipitation                                                                    | PrecAvg     | 651   | 48    | 45    | 49    | 32    | 31    | 40    | 52    | 76    | 66    | 74    | 80    | 58    | (15) |
| (16) |                                                                                  | PrecMax     | 819   | 101   | 85    | 102   | 82    | 72    | 88    | 103   | 133   | 131   | 128   | 195   | 133   | (16) |
| (17) |                                                                                  | PrecMin     | 481   | 5     | 9     | 9     | 8     | 11    | 16    | 9     | 20    | 22    | 34    | 25    | 17    | (17) |
| (18) |                                                                                  | PrecStd     | 89    | 24    | 21    | 24    | 15    | 15    | 20    | 22    | 32    | 28    | 21    | 35    | 29    | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 1.7   | 2.5   | 0.0   | 2.9   | 12.1  | 19.8  | 23.4  | 20.4  | 15.3  | 9.2   | 4.1   | 1.7   | (19) |
| (20) |                                                                                  |             | MCWB  | 0.6   | 1.6   | -1.0  | 0.7   | 6.7   | 12.9  | 18.2  | 15.4  | 12.2  | 6.9   | 2.9   | 0.7   | (20) |
| (21) |                                                                                  | 2%          | DB    | 0.7   | -0.3  | -1.2  | 1.8   | 8.4   | 17.1  | 19.3  | 18.3  | 13.8  | 7.9   | 2.9   | -1.0  | (21) |
| (22) |                                                                                  |             | MCWB  | -0.3  | -1.2  | -2.0  | 0.3   | 5.2   | 11.6  | 14.8  | 14.4  | 11.2  | 6.1   | 1.8   | -1.9  | (22) |
| (23) |                                                                                  | 5%          | DB    | -2.7  | -2.2  | -2.8  | 1.0   | 6.9   | 15.4  | 17.5  | 16.7  | 12.7  | 6.9   | 2.0   | -3.2  | (23) |
| (24) |                                                                                  |             | MCWB  | -3.5  | -3.1  | -3.6  | -0.1  | 4.2   | 10.6  | 13.9  | 13.8  | 10.4  | 5.3   | 0.9   | -4.1  | (24) |
| (25) |                                                                                  | 10%         | DB    | -7.0  | -4.9  | -4.6  | 0.2   | 5.6   | 13.8  | 16.1  | 15.4  | 11.7  | 5.9   | 0.8   | -5.5  | (25) |
| (26) |                                                                                  |             | MCWB  | -7.7  | -5.6  | -5.5  | -0.9  | 3.4   | 9.7   | 12.9  | 12.8  | 9.6   | 4.6   | -0.2  | -6.4  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 0.9   | 1.5   | -0.8  | 1.2   | 7.5   | 14.4  | 18.3  | 16.3  | 12.7  | 7.5   | 3.0   | 0.8   | (27) |
| (28) |                                                                                  |             | MCDWB | 1.5   | 2.4   | -0.2  | 2.2   | 11.4  | 19.1  | 22.7  | 19.2  | 14.7  | 8.6   | 4.0   | 1.6   | (28) |
| (29) |                                                                                  | 2%          | WB    | -0.4  | -1.2  | -1.9  | 0.6   | 5.4   | 12.1  | 15.5  | 14.9  | 11.6  | 6.2   | 1.9   | -1.7  | (29) |
| (30) |                                                                                  |             | MCDWB | 0.7   | -0.3  | -1.1  | 1.5   | 8.0   | 16.0  | 18.6  | 17.5  | 13.5  | 7.6   | 2.7   | -1.1  | (30) |
| (31) |                                                                                  | 5%          | WB    | -3.4  | -3.0  | -3.6  | 0.0   | 4.5   | 10.9  | 14.3  | 13.9  | 10.6  | 5.5   | 1.0   | -4.1  | (31) |
| (32) |                                                                                  |             | MCDWB | -2.7  | -2.1  | -2.8  | 1.0   | 6.6   | 15.0  | 17.0  | 16.3  | 12.4  | 6.8   | 1.9   | -3.2  | (32) |
| (33) |                                                                                  | 10%         | WB    | -7.7  | -5.6  | -5.4  | -0.9  | 3.6   | 10.0  | 13.2  | 13.1  | 9.8   | 4.5   | -0.2  | -6.4  | (33) |
| (34) |                                                                                  |             | MCDWB | -7.0  | -4.9  | -4.7  | 0.2   | 5.5   | 13.2  | 15.7  | 15.2  | 11.4  | 5.8   | 0.8   | -5.5  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 5.0   | 5.2   | 5.5   | 4.6   | 4.6   | 7.1   | 6.2   | 5.6   | 5.3   | 4.7   | 5.3   | 5.1   | (35) |
| (36) |                                                                                  |             | MCDBR | 5.4   | 6.3   | 5.3   | 4.0   | 7.0   | 11.0  | 9.5   | 8.5   | 7.2   | 4.6   | 5.3   | 6.8   | (36) |
| (37) |                                                                                  |             | MCWBR | 5.0   | 6.0   | 5.0   | 3.3   | 5.0   | 7.3   | 6.5   | 6.1   | 5.5   | 4.1   | 5.0   | 6.3   | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 5.2   | 6.2   | 5.2   | 3.8   | 6.4   | 10.3  | 8.7   | 7.7   | 6.1   | 4.2   | 5.1   | 6.6   | (38) |
| (39) |                                                                                  |             | MCWBR | 5.1   | 6.0   | 5.0   | 3.2   | 4.7   | 7.2   | 6.5   | 5.9   | 5.2   | 4.0   | 5.0   | 6.4   | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.226 | 0.272 | 0.332 | 0.398 | 0.449 | 0.411 | 0.446 | 0.432 | 0.343 | 0.317 | 0.260 | 0.209 | (40)  |      |
| (41) |                                                                                  | taud        | 2.052 | 2.106 | 2.008 | 1.890 | 1.936 | 2.243 | 2.174 | 2.189 | 2.443 | 2.289 | 2.093 | 1.959 | (41)  |      |
| (42) |                                                                                  | Ebn at Noon | 521   | 706   | 764   | 774   | 766   | 816   | 771   | 747   | 777   | 649   | 454   | 358   | (42)  |      |
| (43) |                                                                                  | Edh at Noon | 41    | 74    | 116   | 159   | 165   | 124   | 130   | 119   | 77    | 63    | 40    | 28    | (43)  |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.35  | 1.16  | 2.73  | 4.50  | 5.27  | 5.22  | 4.91  | 3.55  | 2.34  | 1.10  | 0.37  | 0.17  | (44)  |      |
| (45) |                                                                                  | RadStd      | 0.05  | 0.09  | 0.27  | 0.43  | 0.51  | 0.59  | 0.51  | 0.34  | 0.19  | 0.13  | 0.05  | 0.02  | (45)  |      |

## Historical Trends

|      |                       | Heating |        | Cooling |       |       | Degree-Days |         |         |         |     |
|------|-----------------------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|-----|
|      |                       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |     |
|      |                       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     | (m) |
| (46) | Station Only          | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     | N/A |
| (47) | Regional (1 neighbor) | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     | N/A |

Nomenclature: See separate page